Twistor–Splitor Diagnostics for Recamán Obstruction Events

A Seeded Gap-Dynamics Approach Using $\{3,2,1\}\{2,1,0\}$

Working dossier

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Abstract

This note records a finished first-stage result for modeling Recamán-style obstruction events. The main conclusion is negative-positive: direct classification of raw event starts, ends, and singletons is too easy under weak controls and is likely contaminated by event geography or range artifacts. The more credible signal appears in the gaps between event starts. We therefore move from raw-integer classification to a seeded event-transition model using the basis $\{3,2,1\}\{2,1,0\}$, local delay embeddings, and a Twistor–Splitor diagnostic. The purpose is not to prove a hidden smooth attractor, but to define a testable geometric language for distinguishing coherent continuation from obstruction-driven splitting.

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1 Core Recamán Object

Let $a_0 = 0$. The classical Recamán update is

$$a_n = \begin{cases} a_{n-1} - n, & a_{n-1} - n > 0 \text{ and } a_{n-1} - n \notin \{a_0, \dots, a_{n-1}\}, \\ a_{n-1} + n, & \text{otherwise.} \end{cases}$$

The operator label is

$$o_n = \begin{cases} D, & \text{down/free step,} \\ U, & \text{up/blocked step.} \end{cases}$$

The obstruction signal is not treated as random noise. It is deterministic and history-dependent: the decision at step n depends on the current value, the index, and the occupied set of previously visited states.

2 Event Compression

The raw obstruction dataset is a large list of singletons and intervals. Instead of expanding every interval into all contained integers, define an event interval

$$I_i = [s_i, e_i], \qquad \ell_i = e_i - s_i + 1,$$

where s_i is the event start, e_i is the event end, and ℓ_i is the event length.

The event-start gap is

$$g_i = s_i - s_{i-1}.$$

The recommended object of study is the transition stream

$$\mathcal{G} = (g_2, g_3, \dots, g_m),$$

not the raw set of integers. The gap sequence is where the strongest nontrivial signal appeared.

3 The Seed $\{3,2,1\}\{2,1,0\}$

Define the seed basis

$$v_1 = (3,2,1)^\top, \qquad v_2 = (2,1,0)^\top,$$

and the seed matrix

$$B = \begin{pmatrix} 3 & 2 \\ 2 & 1 \\ 1 & 0 \end{pmatrix}.$$

For a scalar input x, such as an event start s_i, event length ℓ_i, or gap g_i, define two symbolic counts

$$c_{321}(x) = \#\{d \in \mathrm{digits}(x) : d \in \{3,2,1\}\},$$
$$c_{210}(x) = \#\{d \in \mathrm{digits}(x) : d \in \{2,1,0\}\}.$$

The seed lift is

$$\mathcal{S}_{321|210}(x) = B \begin{pmatrix} c_{321}(x) \\ c_{210}(x) \end{pmatrix}.$$

This seed is not yet a law. It is a low-dimensional symbolic probe. Its value is judged by whether it survives ablation against digit-only and residue-only baselines.

4 Best Current Approach

The best current pipeline is

$$\boxed{\text{raw list} \to \text{event intervals} \to \text{event-start gaps} \to \text{delay/Möbius embedding} \to \text{Twistor–Splitor score}}$$

4.1 Branch Types

Each event is assigned a branch type:

$$b_i = \begin{cases} 0, & \ell_i = 1 \quad \text{singleton}, \\ 1, & \ell_i = 2 \quad \text{pair event}, \\ 2, & 3 \le \ell_i \le 50 \quad \text{short interval}, \\ 3, & 51 \le \ell_i \le 1000 \quad \text{medium interval}, \\ 4, & \ell_i > 1000 \quad \text{huge interval}. \end{cases}$$

Gap contraction and explosion branches are defined by a rolling median:

$$m_i = \mathrm{med}(g_{i-w}, \ldots, g_{i-1}).$$

Then

$$\mathrm{contraction}_i = \mathbf{1}[g_i < m_i],$$
$$\mathrm{explosion}_i = \mathbf{1}[g_i > 2m_i].$$

4.2 Mini-Recamán Residual

Define the gap residual

$$r_i = g_i - i,$$

and the side label

$$h_i = \mathrm{sgn}(g_i - i).$$

A backward-collision proxy can be defined using only previous event intervals:

$$\text{back}_i = s_i - i,$$

$$\text{collide}_i = \mathbf{1}\left[\text{back}_i \in \bigcup_{j<i} I_j\right].$$

The forced-forward proxy is

$$\text{force}_i = \mathbf{1}[\text{back}_i \leq 0 \text{ or } \text{collide}_i = 1].$$

The rule is deliberately causal: no future intervals may be used in this feature.

5 Delay and Möbius-Style Embedding

Given a scalar observable x_i, such as g_i, s_i/n_i, or a_i/i, define a delay vector

$$p_i = (x_i, x_{i-\tau}, x_{i-2\tau}) \in \mathbb{R}^3.$$

Recommended observables include

$$x_i = g_i, \qquad x_i = \log(1 + g_i),$$

$$x_i = \frac{a_i}{i}, \qquad x_i = a_i - a_{i-1},$$

and the seeded gap observable

$$x_i = \mathcal{S}_{321|210}(g_i).$$

The embedding is not claimed to be a guaranteed Takens reconstruction. It is a practical state-reconstruction heuristic for a deterministic, history-dependent integer process.

6 Twistor–Splitor Diagnostic

The visual hypothesis is that local embedded segments show two competing regimes:

- **Twistor sector:** coherent continuation, smooth transport of local direction, ribbon-like flow, low fragmentation.
- **Splitor sector:** obstruction-induced branching, discontinuity, ladder/cage formation, high fragmentation.

Let $p_i \in \mathbb{R}^3$ be embedded points. Define local step vectors

$$u_i = p_i - p_{i-1},$$

and normalized tangents

$$t_i = \frac{u_i}{\|u_i\| + \varepsilon}.$$

6.1 Local Coherence

For a window $W_i = \{i - w, \ldots, i + w\}$, define tangent coherence

$$C_i = \frac{1}{|W_i| - 1} \sum_{j \in W_i} |\langle t_j, t_{j-1} \rangle|.$$

High C_i means local direction is preserved.

6.2 Curvature Smoothness

Define discrete curvature proxy

$$\kappa_i = \|t_i - t_{i-1}\|.$$

Then define curvature stability

$$K_i = \exp\left(-\operatorname{Var}_{j \in W_i}(\kappa_j)\right).$$

High K_i means curvature changes smoothly rather than fragmenting.

6.3 Branching Degree

Build a local k-nearest-neighbor graph on the embedded points. Let d_i be the graph degree of point p_i. Define normalized branching

$$B_i = \frac{d_i - 2}{k - 2 + \varepsilon},$$

clipped to $[0, 1]$. High B_i means the point sits near a locally branching region rather than a simple path.

6.4 Operator Entropy

Let $o_i \in \{D, U\}$ be the local operator label. In a window W_i, define

$$P_i(o) = \frac{1}{|W_i|} \#\{j \in W_i : o_j = o\}.$$

The local operator entropy is

$$H_i = -\sum_{o \in \{D,U\}} P_i(o) \log(P_i(o) + \varepsilon).$$

High entropy means the region mixes down/free and up/blocked operators.

6.5 Fragmentation Score

Let q_i be the number of connected components induced by the local k-NN subgraph on W_i. Define

$$F_i = \frac{q_i - 1}{|W_i| - 1 + \varepsilon}.$$

High F_i means local continuity breaks into separated pieces.

6.6 Twistor Score

The Twistor score is high when direction and curvature are coherent and branching is low:

$$\boxed{\mathsf{Tw}_i = C_i^{\alpha} K_i^{\beta} (1 - B_i)^{\gamma} (1 - F_i)^{\delta}}$$

with default exponents

$$\alpha = \beta = \gamma = \delta = 1.$$

6.7 Splitor Score

The Splitor score is high when branching, fragmentation, and operator entropy are high:

$$\boxed{\mathsf{Sp}_i = B_i^{\lambda} F_i^{\mu} \left(\frac{H_i}{\log 2 + \varepsilon} \right)^{\nu} (1 - C_i)^{\rho}}$$

with default exponents

$$\lambda = \mu = \nu = \rho = 1.$$

6.8 Sector Label

Define the sector label

$$\chi_i = \begin{cases} T, & \mathsf{Tw}_i - \mathsf{Sp}_i > \theta, \\ S, & \mathsf{Sp}_i - \mathsf{Tw}_i > \theta, \\ M, & |\mathsf{Tw}_i - \mathsf{Sp}_i| \leq \theta, \end{cases}$$

where T is a twistor sector, S is a splitor sector, and M is mixed.

7 Validation Targets

The diagnostic is useful only if it improves separation or explanation under strict controls.

7.1 Primary Test

Test whether the sector label predicts operator regime:

$$\chi_i \to o_i.$$

Useful statistics include:

- AUC for U versus D,
- k-NN neighborhood purity,
- operator entropy by sector,
- blocked-step enrichment in splitor sectors,
- free/down-step enrichment in twistor sectors.

7.2 Secondary Test

Test whether gap features outperform raw-integer features:

$$\text{features}(g_i) \quad \text{versus} \quad \text{features}(s_i).$$

The robust result should be stated as:

gap dynamics provide a stronger and cleaner signal than raw event integer classification.

8 Current Empirical Milestone

The current experimental summary is:

Experiment	Mean AUC	Interpretation
Singleton starts	0.9951	suspicious; likely weak-control leakage/geography
Range starts	0.9988	suspicious; likely weak-control leakage/geography
Range ends	0.9975	suspicious; likely weak-control leakage/geography
Gaps between event starts	0.7675	credible and worth pursuing
Dense random seed classifier	0.6633	weak digit/residue shell

The meaningful milestone is not the near-perfect raw classification. The meaningful result is that event-start gaps remain learnable at a nontrivial level.

9 Recommended Control Protocol

For raw starts and ends, replace weak random controls with local decoys:

$$s_i \pm 1, \quad s_i \pm 2, \quad s_i \pm 3, \quad s_i \pm 5, \quad s_i \pm 8, \quad s_i \pm 13,$$

excluding true event points.

If raw-start AUC collapses under local decoys, the previous raw result was geography. If it survives, then local boundary structure exists.

For gap modeling, use blocked validation across event order:

$$\text{train on early event bands, test on later event bands.}$$

This prevents the model from learning only global density bands.

10 Finished Claim

A safe finished claim is:

> The raw obstruction list is too easy to classify under weak controls, but this likely reflects event geography and range artifacts. After compressing the list into event intervals, the gaps between event starts show a more credible signal. The seed $\{3, 2, 1\}\{2, 1, 0\}$ is therefore best treated as a symbolic probe over gap dynamics, not as a direct raw-integer law. The Twistor–Splitor diagnostic provides a local geometric language for separating coherent continuation from obstruction-driven splitting in delay/Möbius-style embeddings.

A Appendix A: Failed or Quarantined Approaches

A.1 A.1 Raw Expanded-Range Classification

Expanding all intervals into all contained integers creates an enormous positive class and makes the model vulnerable to learning density bands, contiguous range artifacts, and magnitude geography. Near-perfect AUCs under weak controls should not be treated as discovery.

A.2 A.2 Dense Random Matrix over 42 Features

A random matrix search over digit counts, residues, last digits, equality flags, and the seed lift produced a mean RF CV AUC around 0.6633. This is weak signal, not noise, but the best rows were dense and hard to interpret. The dominant terms were ordinary digit and residue features:

$$\text{digit counts}, \quad n \bmod p, \quad \text{last digits}, \quad \text{digit-sum flags}.$$

This is classified as a weak digit/residue shell, not a clean obstruction law.

A.3 A.3 Overclaiming Takens-Style Embeddings

Delay embeddings are useful here as reconstruction heuristics, but the Recamán process is discrete, nonstationary, and history-dependent. It is not automatically a smooth compact dynamical system satisfying the assumptions of classical Takens-style reconstruction. The embedding should be used as a feature space, not as proof of a true hidden manifold.

A.4 A.4 Big Neural Networks Too Early

A neural net can maximize matches, but it can also hide leakage and geography. Before neural models, the correct protocol is:

local decoys $\to$ blocked CV $\to$ gap dynamics $\to$ sparse symbolic rules $\to$ only then neural models.

A.5 A.5 Persistent Homology Too Early

Persistent homology may become useful after local sector structure is validated. It is premature if basic label separation, neighbor purity, and boundary concentration are not yet established.

A.6 A.6 Direct Deterministic Seed Law

The seed

$$\{3, 2, 1\}\{2, 1, 0\}$$

should not be claimed as a universal law. It is a compact symbolic basis. Its survival depends on ablation:

$$\text{all features} \quad \text{vs.} \quad \text{all features except seed} \quad \text{vs.} \quad \text{seed only}.$$

If the seed adds no independent value, it remains notation rather than mechanism.

B Appendix B: Minimal Sparse Rule Search

Instead of dense random matrices, use sparse rows:

$$m_j \in \{-2, -1, 0, 1, 2\}^d,$$

with at most 4 to 6 nonzero entries per row. The signature is

$$z_j(x) = \mathbf{1}[m_j^\top x > 0].$$

A row is kept only if it is:

- interpretable,
- stable under blocked CV,
- not destroyed by local decoys,
- not reducible to digit length or magnitude.

C Appendix C: Suggested Next Experiments

1. Re-run raw starts and ends against local decoys.
2. Re-run gap classification under blocked event-order CV.
3. Compute Tw_i, Sp_i, and χ_i on delay embeddings of g_i.
4. Test whether S-sectors enrich blocked/up events.
5. Test whether T-sectors enrich free/down continuation.
6. Ablate the seed lift $\mathcal{S}_{321|210}(g_i)$.
7. Replace dense random matrices with sparse symbolic rules.